SIMBLOCKS // RESULT

2026-10-15 23:25Z

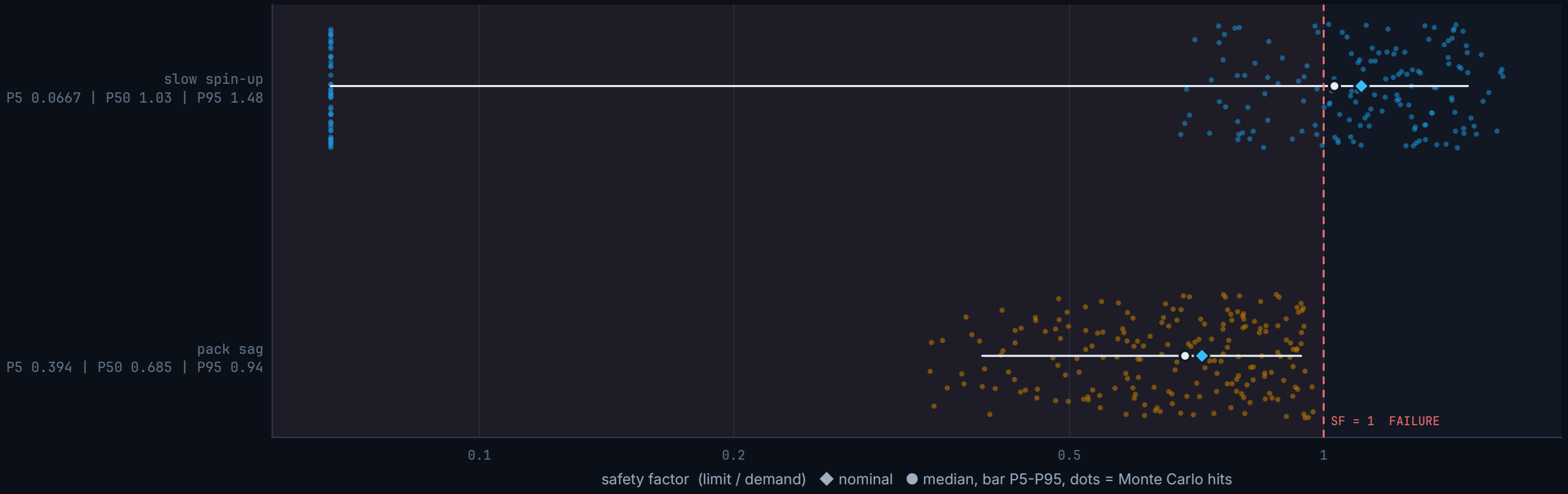
TRIAL 1 WEAPON SPIN-UP (SYSTEM FILE)

● PASS VERIFIED

How fast does the weapon spin up, and does the pack sag below 3.3 V/cell?

SURVIVES?	SLOW SPIN-UP	PACK SAG [GOVERNING]
FAIL	WARN	FAIL
NO	SF 1.11	SF 0.718
slow spin-up SF 1.11, pack sag SF 0.718	P(fail) 46% | SF P5 0.0667	P(fail) 100% | SF P5 0.394

Margin to each user-defined failure mode

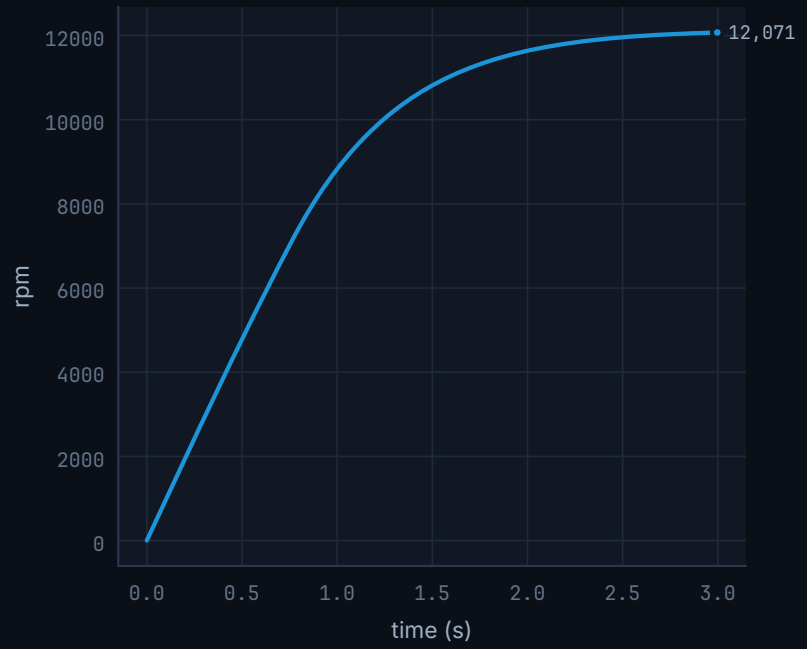


Trial 1 weapon spin-up (system file): VERIFIED | PASS 12 WARN 1 FAIL 0 SKIP 0 | 220 runs, 6.0 s

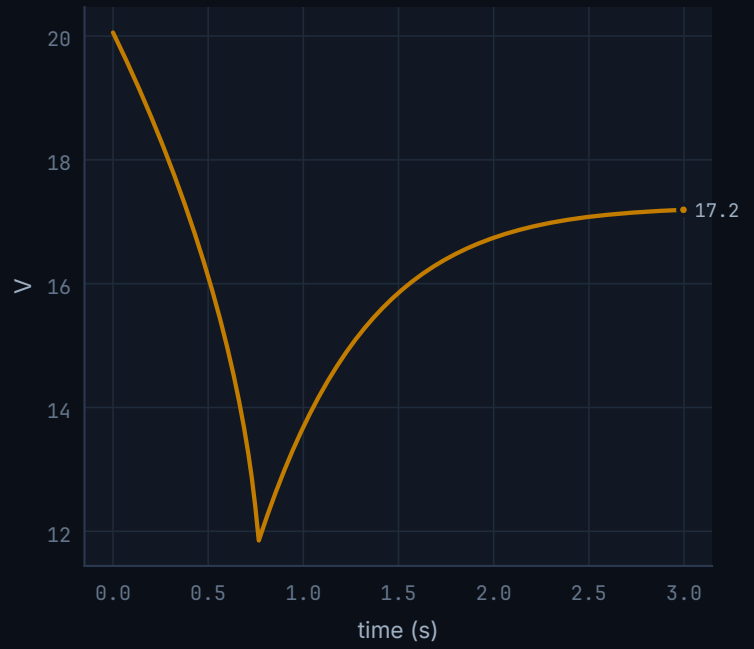
TRIAL 1 WEAPON SPIN-UP (SYSTEM FILE)

Time histories of the governing states (nominal case)

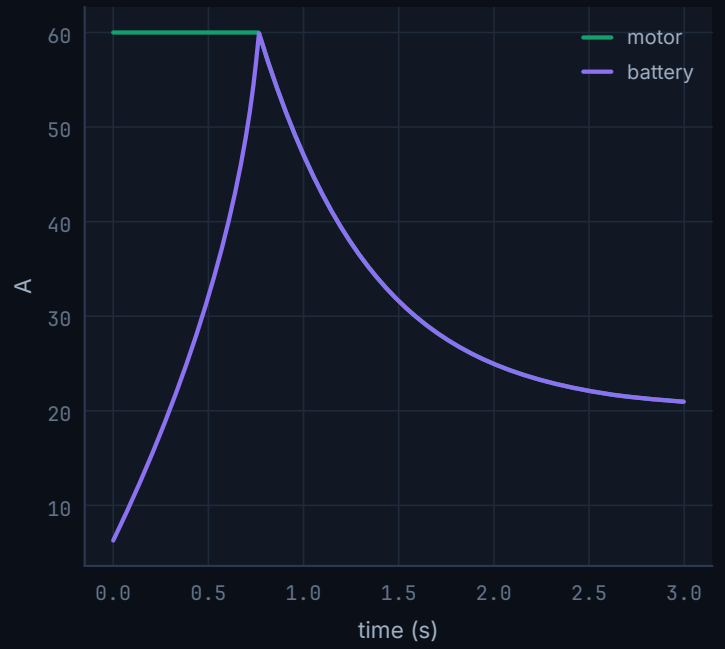
Weapon speed



Pack voltage



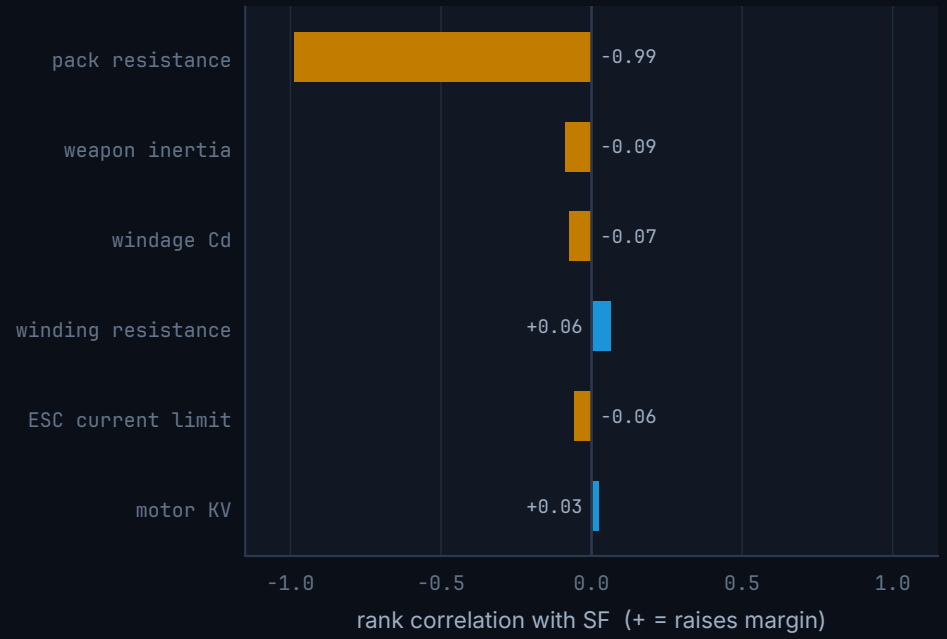
Currents



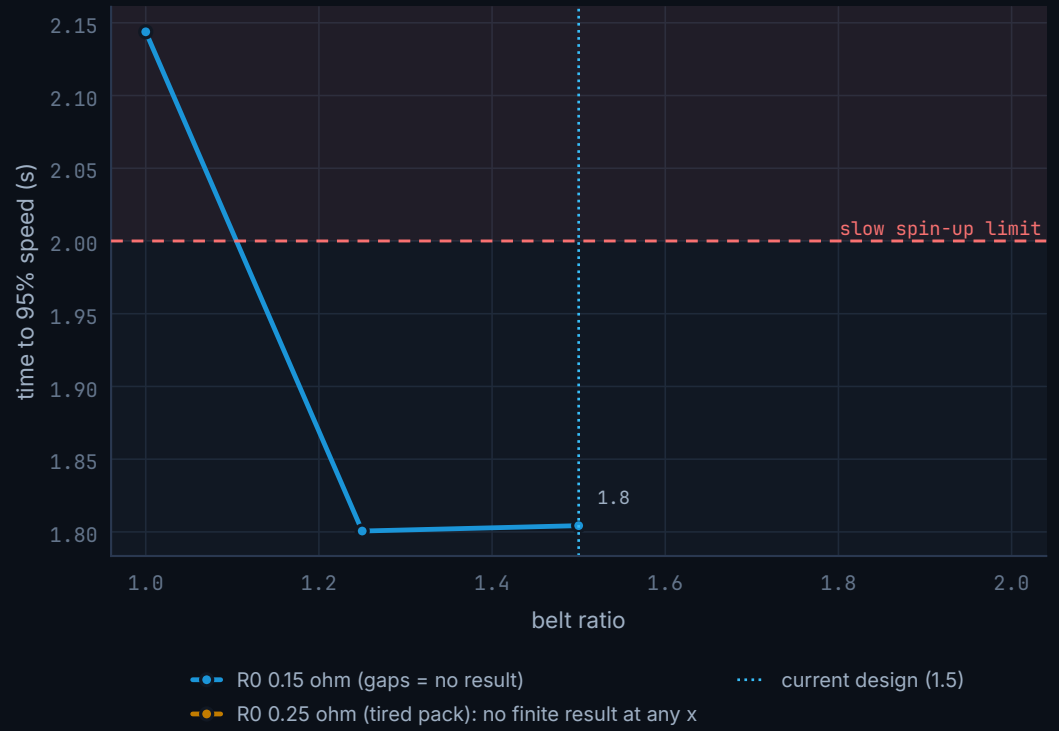
TRIAL 1 WEAPON SPIN-UP (SYSTEM FILE)

What to design around: drivers of the governing margin and the design-variable sweep

Drivers of 'pack sag' margin



t95_s vs belt ratio



SIMBLOCKS // VERIFICATION

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TRIAL 1 WEAPON SPIN-UP (SYSTEM FILE)

● PASS VERIFIED

Automated verification battery

CHECK	STATUS	DETAIL
determinism	● PASS	bit-identical repeat
finite_outputs	● PASS	all outputs finite
physical_bounds	● PASS	3 bounds respected
stability_margin	● PASS	critical dt 0.316 s / dt 0.0001 s = 3156.0x
dt_convergence	● PASS	t95_s: err 0% (order inf); rpm_final: err 0.00037% (order 1); v_bus_min: err 0.0039% (order 3.7); I_peak_A: err 0% (order inf); soc_final: err 0.00...
energy_ledger	● PASS	worst closure residual 0.0073% across 3 dt levels (tol 5%)
limit:ESC limit 0 A → weapon never moves	● PASS	matches expected limit
knockout:no motor	● PASS	t95_s +1.56e+03%, rpm_final -100%, v_bus_min -100%, I_peak_A -100%, soc_final -100%, v_min_per_cell -100%, mAh_used +1.73e+03%
knockout:no windage	● PASS	t95_s -24.4%, rpm_final +20.3%, v_bus_min +0.0764%, I_peak_A +0%, soc_final +1.53%, v_min_per_cell +0.0764%, mAh_used -26.5%
knockout:no bearing	● PASS	t95_s -0.615%, rpm_final +0.129%, v_bus_min +0.00249%, I_peak_A +0%, soc_final +0.0128%, v_min_per_cell +0.00249%, mAh_used -0.222%
sensitivity	● PASS	6 inputs swept low/high (trends reported, not judged)
monte_carlo	● PASS	200 samples, 0 failed finite/bounds/ledger (0.0%)
failure_modes	● WARN	slow spin-up: nominal SF 1.11, P(fail) 46.5%; pack sag: nominal SF 0.718 FAILED, P(fail) 100%